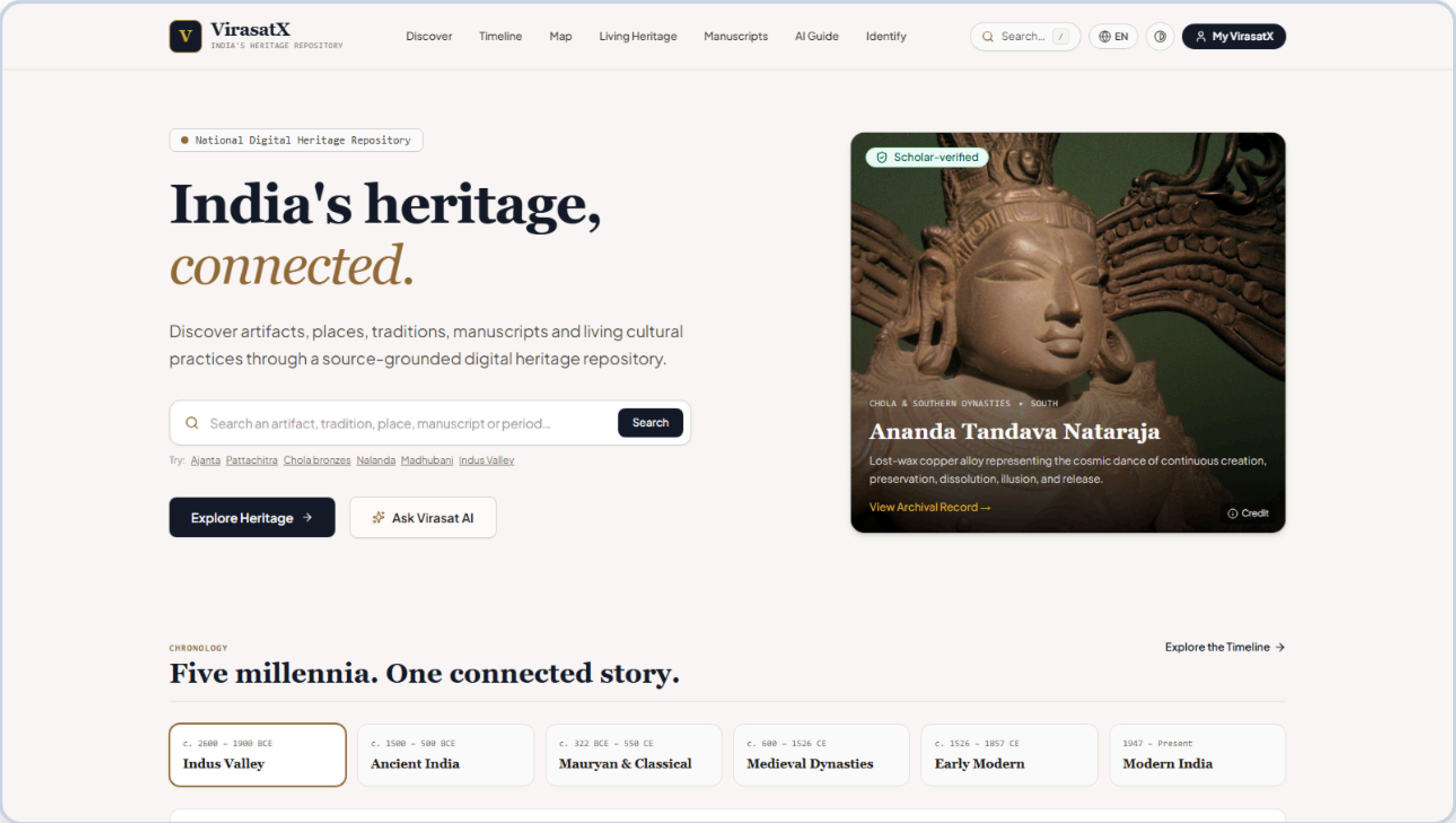
SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –** SIH26197
- **Problem Statement Title :** Student Innovation—Ideas that showcase the rich cultural heritage and traditions of India.
- **Theme :** Heritage & Culture
- **PS Category :** Software
- **Lead Organization :** All India Council for Technical Education (AICTE)
- **Team ID :** [YOUR TEAM ID]
- **Team Name :** VirasatX

ONE-LINE PROJECT PITCH

"An AI-powered digital heritage ecosystem that connects monuments, artifacts, traditions, places, people and knowledge into one discoverable cultural experience."



"India's Heritage, Understood, Preserved and Experienced."

VirasatX
Heritage Ecosystem

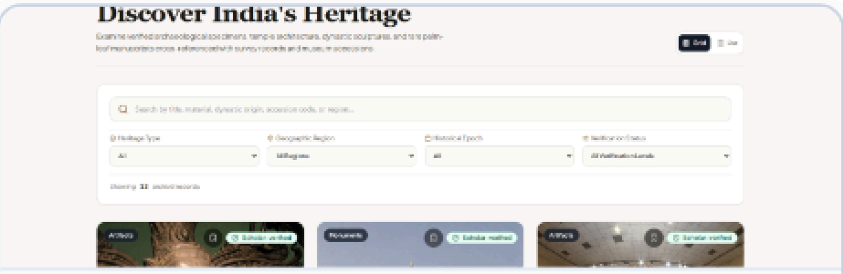
VIRASATX

(India's Heritage, Understood, Preserved & Experienced)

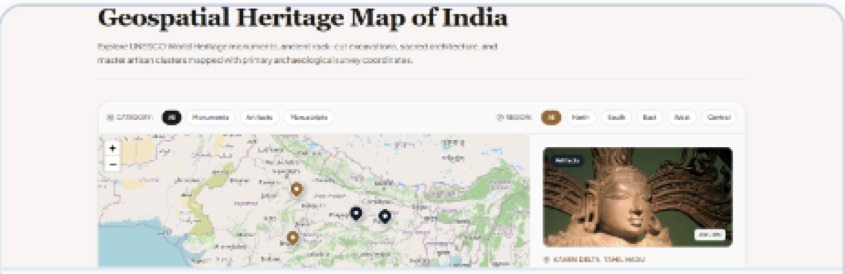
SMART INDIA HACKATHON 2026

PROBLEM EXISTING

- **Fragmented Heritage Knowledge :** India's heritage data is scattered across disconnected monuments, archives, museums, manuscripts, and living communities.
- **Passive Digital Discovery :** Existing portals present heritage as isolated, static information instead of an interconnected cultural experience.
- **Low Discoverability of Living Heritage :** Traditional craft guilds, master artisans, and regional folklore remain disconnected from monuments.
- **Context Gap :** Users discover *what* a heritage site is, but struggle to understand its period, place, tradition, people, and cultural context together.



Faceted Catalog



Heritage Map

Living Traditions

Virasat AI Guide

PROPOSED SOLUTION

- **Unified Heritage Repository :** Explore monuments, artifacts, collections, manuscripts, and living traditions in one place.
- **Multimodal AI Guidance :** Ask Virasat AI for research inquiries and upload photos for visual iconography identification.
- **Contextual Continuity :** Chronological 5,000-year timeline, geospatial GIS corridors, and ancient script paleography.
- **Responsible Cultural Travel :** Carrying-capacity advisories and rural craft linkages to empower local artisan economies.

UVP (UNIQUE VALUE PROPOSITION)

Artifact → **Story** → **Period** → **Place** → **Tradition** → **Community** → **Learning** → **Responsible Experience**

"VirasatX doesn't just display heritage — it connects the cultural context around it."

ARCHITECTURE

★ - Innovation

User / Patron / Student
Text Query • Photo Upload • GIS Map Exploration

↓

VirasatX Digital Heritage Core
Grounded Knowledge Catalog • Local Media Assets

↓

ONLINE MODE	AI RESEARCH	LIVING HERITAGE
Interactive 3D & Atlas	Grounded AI & Vision	GI Artisan Guilds

↓

Grounded Evidence & Citations [ASI-104]
Confidence Ratings • Dynamic Curatorial Narration

↓

Unbroken Cultural Horizon Experience
Explore → Learn → Understand → Preserve

ASI Grounded • No PII Exposure • WCAG 2.1 AA • Open Access

TECHNICAL APPROACH

🚀 Start Here | **Choose Cultural Exploration Pathway** • Visual Specimen Scan • Scholarly Inquiry / Text • Geospatial & 3D Studio

VISION AI PATHWAY
- 📷 Upload Artifact / Monument Photo
- ⚙️ Base64 Preprocessing & Client Sanitization
- 🤖 Gemini 2.5 Flash Vision Multimodal Engine
- 🔍 Identify Mudras, Asanas, Drapery & Dynasty
- 🎯 Output: Classification + Confidence Meter

AI HERITAGE GUIDE (RAG)
- 💬 Natural Query (English, Hindi, Sanskrit)
- 📚 Lexical & Semantic Context Vector Retrieval
- 🏛️ Ground against Curated ASI Accession Records
- ⚡ Serverless Backend API Execution (/api/ai-guide)
- 🎯 Output: Grounded Response with [ASI-104] Citations

GEOSPATIAL & 3D STUDIO
- 🗺️ Leaflet OSM Interactive GPS Coordinate Atlas
- 🧭 6 Thematic Cultural Corridors (Chola, Buddhist, etc.)
- 🎮 Three.js WebGL 360° Tactile Inspection Studio
- 💡 Raking, Amber & Neutral Museum Lighting
- 🎯 Output: Tactile Inspection & Audio Narration

TECH STACK

FRONTEND React 19 • Vite 6	LANGUAGE & CSS TypeScript • Tailwind v4	AI & MULTIMODAL Gemini 2.5 Flash	AI SDK @google/genai
3D ENGINE Three.js (WebGL)	MAPPING GIS Leaflet • OSM	BACKEND & DB Supabase • PostgreSQL	DEPLOYMENT Vercel Serverless

PROTOTYPE VERIFICATION

➤ **Secret Shielding :** Gemini API keys execute strictly in backend serverless functions; zero keys in client bundles.

➤ **Defensive SafeImage :** Shimmer skeletons with zero layout shift (CLS < 0.05) and archival parchment fallback.

Principle : Modular • Scalable • Secure • Evidence-Aware

VirasatX
Project Viability

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

FEASIBILITY :

Technical : Proven Tech Stack — React 19, TypeScript, Vite, Supabase PostgreSQL, and Three.js. Smooth cross-platform web execution with zero heavy native app download requirements.

Secret Protection & Security : Serverless execution ensures Gemini API keys and database credentials are strictly isolated from client-side bundles.

Innovation : Domain-grounded RAG linking inquiries directly to verified ASI accession records (e.g. `[ASI-104]`) with confidence scoring.

Operational : Runs efficiently on low-bandwidth connections through client-side tile caching and progressive `<SafeImage />` fallbacks.

Cultural Data : Uses open cultural access datasets structured to ASI and National Mission for Manuscripts (NMM) archival standards.

VIABILITY :

Educational Reach : Enables schools, rural colleges, and researchers across India to inspect high-resolution monuments, manuscripts, and 3D bronzes remotely.

Policy & National Mission : Directly supports AICTE Student Innovation, Ministry of Culture digitization drives, and UNESCO Sustainable Tourism (SDG 11).

Infrastructure Overhead : Serverless edge deployment and Supabase free-tier database keep initial operational costs near zero, scaling seamlessly with traffic.

Living Heritage Inclusion : Directly spotlights GI-certified master artisan cooperatives (Swamimalai Bronzes, Aranmula Mirrors, Patan Patola) to foster rural craft livelihoods.

Scalability & Expansion : Modular database schema allows continuous ingestion of state archaeological records and regional manuscript repositories.

TECHNICAL CHALLENGES :

Risk : Generative AI models risk hallucinating historical dates, dynastic chronologies, and mudra attributions.

Mitigation : Inquiries are strictly grounded against curated ASI accession catalogs; prompts enforce low temperature (0.2), cite explicit record numbers, and display transparent confidence meters.

CULTURAL & ADOPTION CHALLENGES :

Risk : Traditional artisans and master sthapatis may face privacy intrusion or contact spam from public directories.

Mitigation : Ethical custodianship protocol connects patrons strictly with certified Geographical Indication (GI) cooperative offices with zero exposure of personal telephone numbers or private home addresses.

IMPACTS & BENEFITS

• **Cultural Preservation & Digitization :** Digitally structures and safeguards fragile palm-leaf and birch-bark manuscripts, epigraphical records, and vanishing oral craft knowledge before physical decay occurs.

• **Educational & Youth Engagement :** Replaces passive textbook memorization with tactile 360° WebGL inspection, interactive geospatial timelines, and voice-assisted exploration for schools and universities nationwide.

• **Artisan Livelihood & Sustainable Tourism (SDG 11) :** Connects cultural tourists with GI-certified master artisan guilds and disperses tourist crowds from overburdened monuments to lesser-known heritage clusters.

Feature	Existing Portals	VirasatX
Interconnected Cultural Context	✗	✓
360° WebGL Archival Studio	✗	✓
Visual Iconography AI Scanner	✗	✓
Grounded ASI Accession Citations	✗	✓
Living GI Artisan Guild Linkages	✗	✓
Ancient Manuscript Decipherment	✗	✓
Responsible Travel & Carrying Capacity	✗	✓
100% Local Media SafeImage Pipeline	✗	✓

HOW VIRASATX WORKS ACROSS MODES

MODE 1: EXPLORER (Public & Students)

- **For :** Students, tourists, and casual cultural enthusiasts.
- **Process :** Interactive search → 3D model inspection → Audio narrative → Timeline navigation.
- **Experience :** Fast, zero-registration, engaging 3D visualization.
- **Best For :** Visual learning, monument discovery, school projects.

MODE 2: SCHOLAR & RESEARCHER

- **For :** Historians, archaeologists, and epigraphists.
- **Process :** Ask Virasat AI → Vector retrieval → Accession citations → Script transcription.
- **Experience :** Evidence-aware RAG, confidence scores, Brahmi/Sharada folios.
- **Best For :** Iconographic analysis, academic research, catalog citations.

MODE 3: LIVING HERITAGE & TRAVEL

- **For :** Cultural travelers and craft patrons.
- **Process :** Plan Visit → Dawn/dusk crowd advisories → Discover nearby GI crafts → Cooperative routing.
- **Experience :** Sustainable tourism, direct artisan visibility, SDG 11 compliance.
- **Best For :** Ethical cultural itineraries, craft preservation, rural economy.

RESEARCH AND REFERENCES

- **Archaeological Survey of India (ASI) :** Primary repository of centrally protected monuments, excavation records, and accession catalogs. **Links :** asi.nic.in / NMMA
- **National Mission for Manuscripts (NMM) :** National database of ancient Indian manuscripts, scripts, and preservation guidelines. **Links :** namami.gov.in / IGNCA
- **Geographical Indications Registry of India :** Official documentation for GI-certified traditional craft practices (GI-029 Swamimalai, GI-007 Aranmula). **Link :** ipindia.gov.in
- **UNESCO World Heritage Centre :** Cultural criteria and sustainable tourism framework for monumental conservation (SDG 11). **Link :** whc.unesco.org

SOLUTIONS ALREADY EXIST

- Most existing portals are isolated, static, and text-heavy; designed only for administrative filing.
- Separate websites for monuments, museums, manuscripts, and tourism with zero cross-linking.
- **AVAILABLE SITES :** ASI Portal / NMMA / State Tourism Portals / Commercial Travel Apps

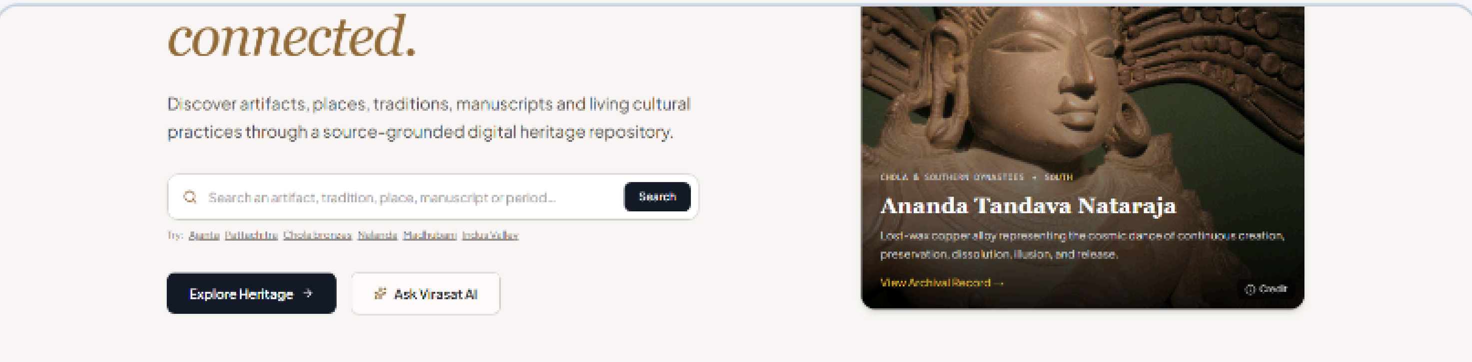
➔

OUR SOLUTIONS STANDS OUT

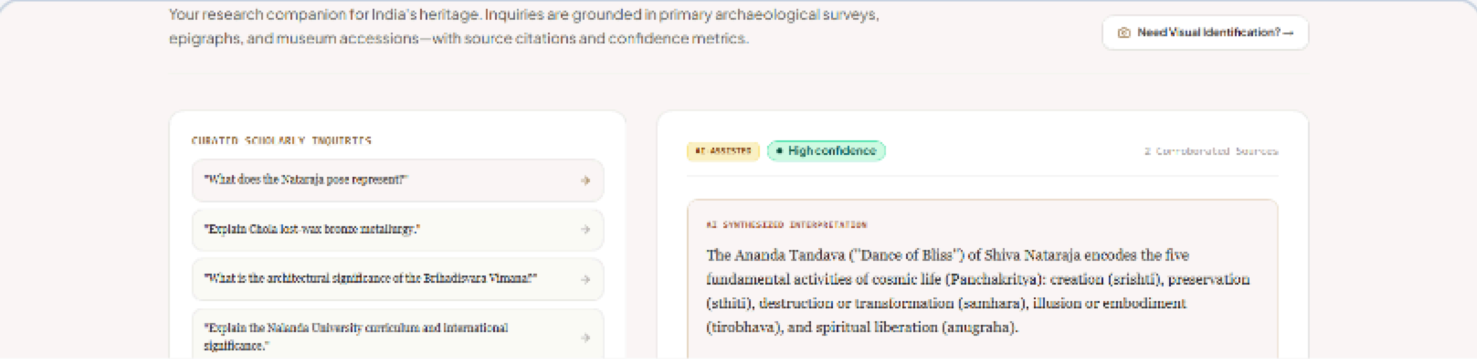
- **One Ecosystem :** Unites monuments, artifacts, manuscripts, and living traditions.
- **Evidence-Aware :** Real-time RAG citing exact ASI accession records [ASI-104].
- **Tactile & Geospatial :** 360° Three.js WebGL studio and 6-corridor Leaflet atlas.
- **Ethical Custodianship :** Promotes GI craft guilds with zero personal contact exposure.

PROJECT RESOURCES

Live Prototype : https://virasatxai.vercel.app/
GitHub Repository : https://github.com/ShrilCarpenter/VirasatX
Full Documentation : Architecture, API Specs & License in README.md
Problem Statement : SIH26197 • Heritage & Culture • AICTE



Interactive Platform



Virasat AI Guide